A proof of Hardin's empirical row recurrences for OEIS A246479 (A246483–A246486), formalized in Lean 4

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Abstract

Let $T(n,k)$ be the number of arrays of length $n+3$ over $\{0,\dots,k\}$ with no two entries among any four consecutive terms summing to k (OEIS A246479 [1]). For fixed n (a "row"), R. H. Hardin recorded linear recurrences in k as "Empirical": explicitly for $n=1,2$ in A246479 and for $n=4,\dots,7$ in A246483–A246486, and by order only $(11,13,\dots,19)$ for $n=3,\dots,7$. We prove that for $1\le n\le 7$ the row is a quasi-polynomial $T(n,k)=A_n(k)+(-1)^k B_n(k)$ with $\deg A_n = n+3$ and $\deg B_n = n$, give A_n and B_n explicitly, and deduce all of these recurrences, which have characteristic polynomial $(1-x)^{n+4}(1+x)^{n+1}$. The proof is machine-checked in Lean 4 with Mathlib [5]: no `sorry`, no `native_decide`, only the axioms `propext`, `Classical.choice`, `Quot.sound`. We conjecture the same structure for every n and check it numerically for $n\le 40$.

1 Introduction and related work

OEIS A246479 (Hardin, 2014) lists "Empirical for row n" recurrences: for $n=1$, $a(k)=3a(k-1)-a(k-2)-5a(k-3)+5a(k-4)+a(k-5)-3a(k-6)+a(k-7)$; for $n=2$ an order-9 recurrence; and "order $11,13,15,17,19$" for $n=3,\dots,7$. Rows 4–7 are the separate entries A246483–A246486, each with an explicit "Empirical" recurrence of order $2n+5$ [2]. For rows 1–3 (A246480–A246482), Colin Barker conjectured quasi-polynomial closed forms in 2018; these, and the column recurrences of A246479, were proved in our earlier work [3], which also supplies the transfer-matrix reduction to eleven triple types used here. No proof of the row recurrences for $n\ge 4$, or of closed forms for these rows, was found in the OEIS or by a web search on October 12, 2026; all of them are marked empirical.

That each row is *some* quasi-polynomial in k is a standard consequence of the transfer-matrix description (the matrix entries are affine in k for each parity of k; cf. Stanley [4], §4.4 and §4.7). The content here is the exact degrees, in particular $\deg B_n = n$ for the alternating part, which is what makes Hardin's recurrences have order exactly $2n+5$, together with explicit formulas and a machine-checked proof.

2 Results

Write P_n^{e} and P_n^{o} for the polynomials with $T(n,k)=P_n^{\mathrm{e}}(k)$ for even k and $T(n,k)=P_n^{\mathrm{o}}(k)$ for odd k, and $A_n=\frac{1}{2}(P_n^{\mathrm{e}}+P_n^{\mathrm{o}})$, $B_n=\frac{1}{2}(P_n^{\mathrm{e}}-P_n^{\mathrm{o}})$.

Theorem 1. *For $n = 4, \dots, 7$ and all $k \geq 0$ (including $k = 0$):*

$$P_4^{\mathrm{e}} = k^7 - 8k^6 + 36k^5 - 97k^4 + 166k^3 - 167k^2 + 76k,$$
$$P_4^{\mathrm{o}} = k^7 - 8k^6 + 36k^5 - 84k^4 + 93k^3 + 16k^2 - 129k + 77,$$
$$P_5^{\mathrm{e}} = k^8 - 10k^7 + 55k^6 - 190k^5 + 442k^4 - 683k^3 + 642k^2 - 277k,$$
$$P_5^{\mathrm{o}} = k^8 - 10k^7 + 55k^6 - 174k^5 + 322k^4 - 256k^3 - 150k^2 + 441k - 227,$$
$$P_6^{\mathrm{e}} = k^9 - 12k^8 + 78k^7 - 327k^6 + 955k^5 - 1969k^4 + 2777k^3 - 2415k^2 + 972k,$$
$$P_6^{\mathrm{o}} = k^9 - 12k^8 + 78k^7 - 308k^6 + 776k^5 - 1144k^4 + 623k^3 + 796k^2 - 1477k + 669,$$
$$P_7^{\mathrm{e}} = k^{10} - 14k^9 + 105k^8 - 516k^7 + 1805k^6 - 4619k^5 + 8596k^4 - 11118k^3 + 8948k^2 - 3356k,$$
$$P_7^{\mathrm{o}} = k^{10} - 14k^9 + 105k^8 - 494k^7 + 1555k^6 - 3206k^5 + 3837k^4 - 1162k^3 - 3525k^2 + 4877k - 1972.$$

For $n = 1, 2, 3$ they are Barker's formulas [3]. In each case $\deg A_n = n + 3$ and $\deg B_n = n$.

Corollary 2. *For $1 \leq n \leq 7$, with $a(k) = T(n, k)$ and $(1-x)^{n+4}(1+x)^{n+1} = \sum_{i=0}^{2n+5} p_i x^i$, we have $\sum_i p_i\, a(k-i) = 0$ for all $k \geq 2n + 5$. This is exactly Hardin's recurrence for $n = 1, 2$ (A246479) and $n = 4, \dots, 7$ (A246483–A246486), coefficient by coefficient, and gives the recurrences of orders $11, \dots, 19$ that A246479 lists for $n = 3, \dots, 7$.*

Proof. $(1 - E^{-1})^{n+4}$ annihilates polynomials of degree $\leq n + 3$, and $(1 + E^{-1})^{n+1}$ annihilates $(-1)^k B(k)$ for $\deg B \leq n$. □

3 Proof of Theorem 1

We use the reduction of [3]: with $\varepsilon = [k \text{ even}]$ there is a vector $V_n \in \mathbb{Z}^{11}$ of type values, $V_0 = \mathbf{1}$, $V_{n+1} = M_{k,\varepsilon} V_n$, and $T(n, k) = F_{k,\varepsilon}(V_n)$, where the entries of M are affine in k and ε and F is a cubic form in k, ε (Lean: `step`, `F`, `T_eq_F`). Fixing $\varepsilon \in \{0, 1\}$, we write out the vectors $V_0, \dots, V_7$ with entries in $\mathbb{Z}[k]$ (`We0`–`We7`, `Wo0`–`Wo7`), check $V_{n+1} = MV_n$ componentwise, and evaluate F. All of these are polynomial identities checked by `ring`. Combining the two parities gives $2T(n, k) = 2A_n(k) + (-1)^k\, 2B_n(k)$ (`row`n`_quasi`), and the recurrences follow by substituting this into both sides (`row`n`_rec`).

Numerical checks. The recursion reproduces all 9999 terms of Hardin's b-file for A246479, and the closed forms were also checked against brute-force enumeration for $n \leq 3$, $k \leq 4$.

Conjecture 3. *For every $n \geq 1$, $\deg A_n = n + 3$ and $\deg B_n = n$; hence row n satisfies the recurrence with characteristic polynomial $(1 - x)^{n+4}(1 + x)^{n+1}$.*

We verified this for $1 \leq n \leq 40$ and $0 \leq k \leq 2n + 29$ (`scripts/check_bfile.py`). The degree bound $\deg A_n \leq n + 3$ is immediate; the content is the bound on B_n. Numerically, every entry of V_L has degree $\leq L$ in k while its dependence on ε has degree $\leq L - 3$, which suggests an inductive proof, but we have not completed one.

4 Formalization

Lean 4 [6], namespace `A246479`. Files `Basic`, `Types`, `Step` are taken unchanged from [3]; `Eps` proves ε for each parity; `Rows` (generated by `scripts/gen/gen_rows.py`) contains, for $n = 1, \dots, 7$: `row`n`_even`, `row`n`_odd` (closed forms), `row`n`_quasi`, and `row`n`_rec` (the recurrence in OEIS form, for all $j \geq 2n + 5$). `Axioms.lean` prints the axioms of all 28 theorems.

5 Reproducibility

- Source: `https://github.com/het4rk/OEIS-A246483-proof`; Lean project in `A246479Rows/`.
- Toolchain `leanprover/lean4:v4.35.0-rc4`; Mathlib `v4.35.0-rc4` (pinned in `lake-manifest.json`).
- Build: `cd A246479Rows; lake exe cache get; lake build; lake env lean Axioms.lean`.
- `verification/` has the build log, axiom output and `VERIFICATION.md`; CI rebuilds on every push, rejects `sorry` and non-standard axioms.

6 Limitations

The formal results cover $n \leq 7$, which are all the rows for which the OEIS states a recurrence; the general-n statement is a conjecture supported by computation. The proof is a direct computation on top of the transfer-matrix reduction of [3]; its value is in turning these empirical formulas into theorems with explicit closed forms and a machine-checked proof.

References

[1] R. H. Hardin, R. Israel, *A246479*, The On-Line Encyclopedia of Integer Sequences, 2014, `https://oeis.org/A246479`.

[2] R. H. Hardin, *A246483–A246486* (rows 4–7 of A246479), The On-Line Encyclopedia of Integer Sequences, 2014, `https://oeis.org/A246483`.

[3] H. Shah (AI-assisted by Grok), *A proof of the conjectured column recurrences for OEIS A246479, formalized in Lean 4*, 2026, `https://github.com/het4rk/OEIS-A246479-proof`.

[4] R. P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge University Press, 2012.

[5] The mathlib Community, *The Lean mathematical library*, in Proc. CPP 2020, ACM, 367–381.

[6] L. de Moura, S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, LNCS 12699, Springer, 2021, 625–635.